The supplemental materials for paper: A Bio-Inspired Energy Cell Framework for Developing Flexible and Resilient Urban Energy Systems

I. A Comprehensive Comparison Among the EC And Related Concepts.

Table S1. The comparison between the proposed EC framework and exiting architecture.

Architecture	Basic Unit	Covered Sectors	Boundary Characteristic	Operational Mode	Typical Application Scenario
Energy Hub	Energy-coupling devices	Multi-energy system	Single-site, without an explicit inter-unit boundary	Device-level dispatch	Optimal multi-energy dispatch at a single site
Microgrid	Local grid	Mainly power system	Relatively fixed, usually defined by physical connection or islanding capability	Islanded/grid-connected operation	Local energy autonomy and islanded operation
Cellular Grid	Grid cell	Power system	Fixed, partitioned by operational boundaries	Distributed coordination	Voltage and frequency control
Urban Cell	Functionalized urban block	Energy-society system	Fixed, partitioned by urban blocks	Planning-oriented	Urban planning and design
Proposed EC framework	Energy cell	Multi-energy system	Flexible, with dynamic aggregation into EC clusters	Hierarchical coordination across EC-cluster-UES	Economic operation, flexibility utilization, and resilience enhancement

For the Energy Hub, its main focus is the modeling and optimal dispatch of multi-energy conversion devices at a single site. It is highly useful for describing the input-output relationship among different energy carriers, but it does not explicitly define inter-unit organization, flexible boundary interaction, or dynamic aggregation among multiple local energy units. By contrast, the EC is not only an energy-conversion node, but a distributed multi-energy functional unit embedded in a wider UES. It integrates distributed generation, loads, storage, conversion devices, and flexible resources within a local region, and further participates in cluster-level and system-level coordination.

For the Microgrid, its core contribution lies in local power-system autonomy, especially islanded/grid-connected operation within a relatively fixed electrical boundary. However, a conventional microgrid is usually electricity-centered, and its boundary is commonly determined by physical connection, ownership, or islanding capability. By contrast, the EC framework emphasizes multi-energy self-balancing and flexible boundary interaction. In other words, an EC is not only a local electrical subsystem, but a multi-energy unit that coordinates electricity, heat, gas, hydrogen, storage, and flexible demand to improve local balancing and support neighboring cells when needed.

For the Cellular Grid, the cell-based idea provides an important reference for distributed coordination in the power sector, especially for voltage and frequency control. However, its scope is mainly limited to the power system, and its cells are generally defined by relatively fixed operational boundaries. The proposed EC framework extends this cell-based organizational logic from the power sector to distributed multi-energy UES. More importantly, ECs are allowed to dynamically aggregate into EC clusters under disturbances, which enables cross-boundary resource reorganization and resilience enhancement.

For the Urban Cell, its contribution lies in extending the energy hub concept to sector and spatial coupling in urban planning. It provides a useful planning-oriented modular viewpoint for cities. However, the Urban Cell is mainly concerned with urban planning and spatial design, whereas the proposed EC framework is more operation- and coordination-oriented. The EC framework focuses on how local multi-energy units can self-balance, interact, and aggregate in real-time or near-real-time operation to support flexibility utilization and resilience enhancement.

Therefore, the core advantage and irreplaceability of the proposed EC framework lie in its integrated consideration of localized multi-energy self-balancing, flexible inter-cell interaction, and dynamic aggregation within an EC-cluster-UES hierarchy. These features distinguish EC from existing mature schemes that usually focus on only one part of this problem, such as single-site energy conversion, fixed-boundary local autonomy, or power-sector distributed coordination.

II. Biological Term Explanations

A. A detailed explanation of Fig.1

In Fig.1, at the component level, representative biological organelles are mapped to key energy-system components: the nucleus corresponds to the energy management center, the vacuole to energy storage, the chloroplast to energy generation, and the mitochondrion to energy utilization. This level emphasizes the functional correspondence between the internal constitution of a biological cell and that of an EC. Specifically,

- Cell chloroplast and EC generation-type units: The compatibility between cell chloroplast and generation-type units in EC is not merely that both are related to energy production, but that both serve as the interface through which external energy is captured and converted into a usable internal energy form. In biological cells, the chloroplast transforms environmental energy into usable biochemical energy; similarly, generation-type units in the EC, especially renewable generation units, convert external primary energy into usable energy.
- Cell vacuole and EC storage-type units: Both cell vacuole and EC storage-type unit have role of buffering, temporary storage, and internal regulation. A vacuole is an internal regulator that helps maintain the stability of the cell. Likewise, storage-type units in the EC are buffering components that shift energy across time, absorb fluctuations, and support local self-balancing.
- Cell mitochondrion and EC consumption-type units: This analogy can be understood from the perspective of functional endpoint of energy utilization. The mitochondrion is where energy substrates (such as the glucose) are processed into ATPs to sustain the activity of the cell; similarly, consumption-type units and local loads in the EC represent the part of the system where energy is ultimately utilized to maintain local services and functions.

At the functional level, the analogy is further extended from component correspondence to system-operating principles.

- The hierarchical organization from cell to organ/tissue to organism-level system is used to motivate the EC-cluster-UES layered structure, highlighting modular aggregation from local units to the whole system.
- The signal interaction among biological cells, such as bioelectrical and synaptic communication, is used to inspire inter-EC information exchange, which may be realized through discrete pulse-like communication rather than continuous global broadcasting.
- The biological regulation and immune-response mechanisms activated under virus invasion or abnormal cells, involving hormones, T-cells, B-cells, are used to analogize the cooperative suppression of uncertainty and disturbances in the EC framework, where neighboring ECs can provide coordinated support through storage and other flexible resources.
- The adaptive metabolic switching of cells under low-oxygen conditions, where ATP production shifts from the mitochondrion-dominated aerobic pathway to glycolysis, is used to analogize adaptive energy substitution in ECs, such as the transition from photovoltaic generation to energy-storage discharge under low solar-radiation conditions.

B. Quick reference index for biological terms & engineering mechanisms.

Table S2. Quick reference index for biological terms & engineering mechanisms.

Biological Item	Biological Meaning	Engineering Mechanism	Functional Similarity
Organismic Cell	The basic functional unit of a living organism, capable of maintaining essential functions through internal regulation	Energy Cell, as the fundamental structural and functional unit of UES, integrating distributed generation, loads, storage, and flexible resources	Both are basic functional units that sustain local operation while contributing to the whole system
Cell Division	The formation of multiple cells from a stem cell	Partitioning the UES into multiple initial ECs based on multi-energy balance and resource allocation	Both create manageable local units from a larger system to support organized operation
Cell Differentiation	Cells specialize into different types according to their roles and characteristics	Functional specialization of ECs according to physical roles, resource endowment, and energy characteristics	Both enable heterogeneous units to perform designated functions within a coordinated system
Cell Nucleus	The control center that regulates cellular behavior and decision-making	Control layer of EC for local energy management and decision-making based on sensed states and interaction signals	Both serve as the decision core of the local unit

Cell Cytoplasm	The internal medium containing functional components that support cellular activity	Physical layer of EC, including energy devices, storage, and aggregated energy facilities	Both contain the physical components that realize the unit's actual functions
Cell Membrane	The interface that separates the cell from its environment and regulates exchanges	Interface layer of EC, supporting standardized information exchange and boundary interaction with neighboring ECs or upper-level operators	Both define boundaries and regulate interaction with the outside environment
Cell-Tissue-System	Multi-level biological organization from local units to larger coordinated structures	EC-cluster-UES hierarchy from local ECs to EC clusters and the whole urban energy system	Both represent hierarchical organization from basic units to system-level coordination
Hormone Regulation	Biological regulation through signaling substances that guide responses across scales	Hierarchical coordination signals used to guide EC scheduling and cross-layer adjustment	Both provide regulation to guide the function adjustment
Reference Hormones	Guidance signals under normal conditions for coordinated regulation	Day-ahead or medium-term reference signals sent from upper levels to guide EC scheduling plans	Both guide local behavior toward a broader coordinated objective under normal conditions
Emergency Hormones	Urgent biological signals released when abnormal conditions are detected	Emergency requests or imbalance signals sent from ECs to neighbors or clusters when local balancing is insufficient	Both trigger additional support only when local regulation is no longer enough
Cell chloroplast	Organelles responsible for energy production	Generation-type flexible resources within EC	Both are associated with energy production capability
Cell Vacuole	Organelles used for storage and regulation of materials	Storage-type flexible resources within EC	Both provide storage and buffering capability for local regulation
Cell Mitochondrion	Organelles responsible for energy consumption and metabolic conversion	Consumption-type flexible demand within EC	Both are associated with energy use and conversion for sustaining local function
Stress Reactivity	Short-term ability to respond and recover from sudden disruptions	Short-term flexibility response capability of resources/ECs/clusters	Both describe fast response to immediate disturbances
Defensive Capability	Mid-term ability to resist risks and take preventive measures	Mid-term capability of ECs/clusters to resist foreseeable operational risks	Both reflect protective capacity against anticipated threats
Immunization	Long-term adaptability and resistance to repeated or evolving threats	Long-term sustainability and adaptability of flexibility planning against future operational changes	Both emphasize enduring protection and adaptation over long horizons
Viral Invasion	External attack that disturbs normal cellular function	Extreme disturbances such as natural disasters	Both represent external disturbance threatening normal system function
Autophagy	Controlled self-reduction of internal activity to maintain survival under stress	Proactive shedding of non-critical loads	Both sacrifice part of local function to preserve the survival of the larger unit
Apoptosis	Programmed removal of damaged cells to protect the organism	Proactive disconnection of an EC from neighboring ECs to isolate fault propagation	Both isolate damaged parts to prevent wider system harm
Paracrine Signaling	Local signaling among nearby cells to coordinate neighboring responses	Parallel coordinated signaling among neighboring EC clusters during post-disaster recovery	Both enable local cooperative recovery through nearby interaction
Synaptic Pulses	Discrete signals transmitted only when bioelectrical thresholds are reached	Event-triggered, encoded discrete communication among ECs under limited bandwidth conditions	Both reduce communication burden by transmitting only significant state changes

III. Modeling Details

A. Guidance on general modeling template of EC-clusters-USE hierarchy

1) What is optimized at each layer:

At the UES layer, the objective is to determine the system-level coordination target for the whole UES, such as the overall operating cost, global energy-accessibility objective, or other system-wide performance indicators. This layer does not optimize detailed local device schedules; instead, it coordinates cluster-level targets from a global perspective. A minimal formulation can be written as:

$$\min_{y_c, z} \sum_{c \in \mathcal{C}} F_c\left(y_c\right) + G\left(z\right)$$
$$\text{s.t.} \begin{cases} \mathcal{A}_c\left(z, y_c\right) = 0, \forall c \in \mathcal{C} \\ \mathcal{B}_c\left(z, y_c\right) \le 0, \forall c \in \mathcal{C} \end{cases} \tag{1}$$

where, y_c denotes the coordination target assigned to cluster c and z is the global decision or coordination variables, $F_c(\cdot)$ represents the cluster-level operating metrics of cluster c, and $G(\cdot)$ captures the system-level objective, $\mathcal{A}_c$ and $\mathcal{B}_c$ are the aggregated feasible region coefficients reported by cluster c, $\mathcal{C}$ is the clusters set.

At the EC-cluster layer, the objective is to translate the upper-layer cluster target into regional coordination references for ECs in this cluster. In other words, this layer determines how the cluster-level target should be allocated among neighboring ECs in a coordinated and implementable manner. A minimal formulation can be written as:

$$\min_{r_i, e_c} \sum_{i \in \mathcal{E}_c} f_i(r_i) + h_i(e_i)$$
$$\text{s.t.} \begin{cases} B_c(r_i, e_i) = y_c \\ \mathcal{M}_i(r_i, e_i) = 0, \forall i \in \mathcal{E}_c \\ \mathcal{N}_i(r_i, e_i) \le 0, \forall i \in \mathcal{E}_c \end{cases} \tag{2}$$

where e_i is the coordination instruction for each EC, r_i is the reference signal for EC i, $f_i(\cdot)$ is the local EC-level operating metrics, $h_i(\cdot)$ is the coordination discipline. $B_c(\cdot)$ maps the EC reference signal and coordinated decisions to the cluster-level decision y_c, $\mathcal{M}_i$ and $\mathcal{N}_i$ are the aggregated feasible region coefficients reported by the i-th EC.

At the EC layer, each EC optimizes its local multi-energy scheduling decisions, including local generation, storage, conversion devices, and controllable demand response, subject to local physical and operational constraints. A minimal formulation can be written as:

$$\min_{x_i} f_i(x_i)$$
$$\text{s.t.} \begin{cases} D_i(r_i, e_i) = x_i \\ g_i(x_i) = 0 \\ q_i(x_i) \le 0 \end{cases} \tag{3}$$

where x_i denotes the local dispatch decision of i-th EC, and feasible operating region is consisted of local equality constraints $g_i(\cdot)$ and inequality constraints $q_i(\cdot)$, $D_i(\cdot)$ maps the EC local dispatch into the reference signal obtained from upper cluster.

Therefore, the hierarchy can be understood as follows: the UES layer determines the global coordination target, the cluster layer allocates this target into regional coordination references, and the EC layer computes physically implementable local schedules.

2) What variables are exchanged across layers

From the UES layer to the EC-cluster layer, the exchanged variables are the cluster-level coordination targets y_c, which specify the desired aggregate operating or exchange pattern of each cluster from the system-level perspective.

From the EC-cluster layer to the EC layer, the exchanged variables are the EC-level coordination references r_i. These references indicate how each EC is expected to contribute to the cluster-level target, for example the target boundary exchanges, balancing references, flexibility expectations, or other coordination-oriented signals.

From the EC layer back to the EC-cluster layer and the EC-cluster layer back to UES-layer, the exchanged information is the aggregated feasible region. This information represents the feasible support capability or operational boundary that each EC/cluster can provide after considering its internal physical constraints.

3) How local and inter-layer feasibility are maintained:

In a single EC, its local feasibility is guaranteed through the constraints $g_i(x_i) = 0$ and $q_i(x_i) \le 0$ in (3). These constraints ensure that each EC schedules its local resources within a physically meaningful and operationally admissible region, including multi-energy balance requirements, device limits, conversion relationships, and local network restrictions.

The inter-layer feasibility between EC-layer and cluster-lay is guaranteed by $\mathcal{M}_i(r_i, e_i) = 0$ and $\mathcal{N}_i(r_i, e_i) \le 0$ in (2), which ensure that the reference signal and coordination instruction of each EC are endogenously determined by its local feasible dispatch, rather than being arbitrarily assigned by the cluster. Therefore, the EC can compute a feasible local action that matches the target set by cluster, make sure the constraint $D_i(r_i, e_i) = x_i$ in (3) is always satisfied.

The inter-layer feasibility between EC-layer and cluster-lay is guaranteed by $\mathcal{A}_c(z, y_c) = 0$ and $\mathcal{B}_c(z, y_c) \le 0$ in (2). Under these feasible-region constraints, the target assigned to each cluster can be physically achieved, ensuring that $B_c(r_i, e_i) = y_c$ always holds and that the cluster-level references assigned to individual ECs are collectively consistent with the target determined by the upper UES layer. In this way, the global coordination objective is connected to lower-layer implementation through a structured aggregation mechanism.

It should be noted that, the added formalization is not intended to provide a single fully general model for all possible UES configurations, which would be difficult to achieve within the scope of a perspective article due to the large diversity of physical infrastructures, energy carriers, and operational objectives. Instead, what we provide is a generic but implementable modeling template for the proposed hierarchy. More specific models can then be developed from this template for particular application scenarios.

B. Detailed decomposition approach and solution framework

To illustrate the decomposition process more clearly, we take the EC-cluster layer as a representative example. At this layer, the coordination problem (2) previous aforementioned can be reformulated into a compact form as follows:

$$\left.\begin{array}{l}\min\limits_{r_i,e_c}\sum\limits_{i\in\mathcal{E}_c} f_i\left(r_i\right)+h_i\left(e_i\right)\\ \text{s.t.}\begin{cases}B_c\left(r_i,e_i\right)=y_c\\ \mathcal{M}_i\left(r_i,e_i\right)=0,\forall i\in\mathcal{E}_c\\ \mathcal{N}_i\left(r_i,e_i\right)\le 0,\forall i\in\mathcal{E}_c\end{cases}\end{array}\right\}\xrightarrow[\text{Compact Form}]{\text{Linear Assumption}}\begin{cases}\min\limits_{X_i,\hat{X}_i}\sum\limits_{i\in\mathcal{E}_c} f_i\left(\boldsymbol{X}_i\right)\\ \text{s.t.}\begin{cases}\mathbb{A}_i\boldsymbol{X}_i-\mathbb{B}_i\hat{\boldsymbol{X}}_i=0\\ \forall i\in\mathcal{E}\end{cases}\end{cases} \tag{4}$$

where $\boldsymbol{X}_i$ is the local EC related variables, and $\hat{\boldsymbol{X}}_i$ is the auxiliary variable introduced to represent the coordinated information exchanged within the cluster and the derived variables from the constraints, $\mathbb{A}$ and $\mathbb{B}$ are the related coefficient matrices. Under this reformulation, the original cluster-level coordination problem can be decomposed into multiple local EC subproblems together with a cluster-level auxiliary-variable update. This reformulation is always feasible because the inter-layer consistency constraints $B_c\left(r_i,e_c\right)=y_c$ and $D_i(r_i,e_i)=x_i$ ensure the reference signal for each EC can be meet by local EC and this reference signal can meet the upper layer target. And the objective of the cluster layer is naturally consisted of local EC objectives, therefore the decomposition is logically.

Specifically, the decomposed subproblems for each EC can be distributed solved through an iterative procedure, which consists of four main steps (take k-th iteration as an example):

- local decision-variable update, where each EC independently solves its own subproblem for X_i

$$\boldsymbol{X}_i^{k+1}=\arg\min_{x_i^k} f_i(\boldsymbol{X}_i)+\frac{\tau_i^k(\boldsymbol{X}_i)}{2}\left\|\boldsymbol{X}_i^k-\hat{\boldsymbol{X}}_i^k+\frac{\boldsymbol{\lambda}_i^k}{\tau_i^k(\boldsymbol{X}_i)}\right\|_2^2 \tag{5}$$

- auxiliary-variable update, where the cluster-level shared variable $\hat{X}_i$ is updated to enforce coordination consistency

$$\hat{\boldsymbol{X}}_i^{k+1}=\arg\min_{\hat{x}^k}\frac{\tau_i^k(\boldsymbol{X}_i)}{2}\sum_{i\in\mathcal{E}_c}\left\|\boldsymbol{X}_i^{k+1}-\hat{\boldsymbol{X}}_i^k+\frac{\boldsymbol{\lambda}_i^k}{\tau_i^k(\boldsymbol{X}_i)}\right\|_2^2 \tag{6}$$

- dual-variable update, where the Lagrange multiplier λ_i associated with the coupling constraint is updated

$$\boldsymbol{\lambda}_i^{k+1}=\boldsymbol{\lambda}_i^k+\tau_i^k(\boldsymbol{X}_i)\left(\boldsymbol{X}_i^{k+1}-\hat{\boldsymbol{X}}_i^{k+1}\right) \tag{7}$$

- adaptive penalty update, where the penalty parameter τ_i is adaptively adjusted to improve convergence.

$$\tau_i^{k+1}(\boldsymbol{X}_i^{k+1})=g\left(\tau_i^k(\boldsymbol{X}_i)\right) \tag{8}$$

Repeating these steps, the local EC decisions, exchanged coordination variables, and multiplier information can iteratively update until the primal and dual residuals satisfy the stopping criterion.

B. Detailed Models in the case studies.

The compact form:

$$\begin{aligned}&\min_{\boldsymbol{x},\boldsymbol{y}}\sum_{t=1}^{T}\boldsymbol{c}^\top\boldsymbol{x}+\lambda\boldsymbol{y}\\ &\text{s.t.}\mathcal{A}_t\boldsymbol{x}=\mathcal{B}_t,\forall t\in T\\ &\quad\mathbf{A_t}\boldsymbol{x}\le\mathbf{b_t},\forall t\in T\\ &\quad\boldsymbol{x}\in\mathcal{X}\end{aligned} \tag{9}$$

The objective function:

$$\sum_{t=1}^{T}c_B^\top P_t^B+c_*^\top P_t^*+\lambda^\top P_t^{\text{EC}} \tag{10}$$

where the cost coefficient vectors c_B,c_* correspond to the energy purchase prices and operational costs, λ is the penalty coefficient. The energy purchases P_t^B including power, gas, heat, and hydrogen purchases. The scheduled output vector P_t^* collectively represent the decision variables of all energy devices, and inter-ECs power exchange vector P_t^{EC}.

The energy balance constraints:

$$P_t^B + \sum_{i=1}^{N_{\mathrm{CHP}}} P_{i,t}^{\mathrm{CHP}} + \sum_{i=1}^{N_{\mathrm{WG}}} P_{i,t}^{f} + \sum_{i=1}^{N_{\mathrm{BE}}} P_{i,t}^{\mathrm{BE,D}} + \sum_{i=1}^{N_{\mathrm{FC}}} P_{i,t}^{\mathrm{FC}} = \sum_{i=1}^{N_D} P_{i,t}^{D} + \sum_{i=1}^{N_{\mathrm{EB}}} P_{i,t}^{\mathrm{EB}} + \sum_{i=1}^{N_{\mathrm{BE}}} P_{i,t}^{\mathrm{BE,C}} + \sum_{i=1}^{N_{\mathrm{EL}}} P_{i,t}^{\mathrm{EL}} \tag{11a}$$

$$\sum_{i=1}^{N_{\mathrm{CHP}}} Q_{i,t}^{\mathrm{CHP}} = \sum_{i=1}^{N_D} Q_{i,t}^{D} \tag{11b}$$

$$\sum_{i=1}^{N_{\mathrm{CHP}}} G_{i,t}^{\mathrm{CHP}} + G_t^D = G_t^{\mathrm{buy}} \tag{11c}$$

$$\sum_{i=1}^{N_{\mathrm{CHP}}} H_{i,t}^{\mathrm{CHP}} + \sum_{i=1}^{N_{\mathrm{EB}}} H_{i,t}^{\mathrm{EB}} + \sum_{i=1}^{N_{\mathrm{FC}}} H_{i,t}^{\mathrm{FC}} = H_t^D \tag{11d}$$

$$M_t^{\mathrm{buy}} + \sum_{i=1}^{N_{\mathrm{EL}}} M_{i,t}^{\mathrm{EL}} + \sum_{i=1}^{N_{\mathrm{HE}}} M_{i,t}^{\mathrm{HE,D}} = \sum_{i=1}^{N_{\mathrm{FC}}} M_{i,t}^{\mathrm{FC}} + \sum_{i=1}^{N_{\mathrm{HE}}} M_{i,t}^{\mathrm{HE,C}} \tag{11f}$$

where Eq. (11a) ensures the active power balance, while Eq. (11b)- Eq. (11e) enforce the balance constraints for reactive power, gas, heat, and hydrogen, respectively. P_t^B the purchased power, $P_{i,t}^{\mathrm{CHP}}$ the power output of CHP, $P_{i,t}^{f}$ the power output of wind generation, $P_{i,t}^{\mathrm{BE,D}}$ the discharging power of energy storage, $P_{i,t}^{\mathrm{FC}}$ the power output of fuel cell (FC), $P_{i,t}^{D}$ the power demand, $P_{i,t}^{\mathrm{EB}}$ the power consumption of electric boiler (EB), $P_{i,t}^{\mathrm{BE,C}}$ the charging power of energy storage, $P_{i,t}^{\mathrm{EL}}$ the power consumption of electrolyzer (EL). $Q_{i,t}^{\mathrm{CHP}}$ the reactive power output of CHP, $Q_{i,t}^{D}$ the reactive power demand, $G_{i,t}^{\mathrm{CHP}}$ the gas consumption of CHP, G_t^D the gas demand, G_t^{buy} the purchased gas. $H_{i,t}^{\mathrm{CHP}}$ the heat output of CHP, $H_{i,t}^{\mathrm{EB}}$ the heat output of EB, $H_{i,t}^{\mathrm{FC}}$ the heat output of FC, M_t^{buy} the purchased hydrogen, $M_{i,t}^{\mathrm{EL}}$ the hydrogen output of EL, $M_{i,t}^{\mathrm{HE,D}}$ the discharging flow of hydrogen storage, $M_{i,t}^{\mathrm{FC}}$ the hydrogen consumption of FC, $M_{i,t}^{\mathrm{HE,C}}$ the charging flow of hydrogen storage.

Device constraints of CHP:

$$P_t^{\mathrm{CHP}} = \left(\eta_{\mathrm{CHP}}^{\mathrm{P}} \rho_{\mathrm{G2P}}\right) G_t^{\mathrm{CHP}} \tag{12a}$$

$$H_t^{\mathrm{CHP}} = \left(\eta_{\mathrm{CHP}}^{\mathrm{H}} \rho_{\mathrm{G2P}}\right) G_t^{\mathrm{CHP}} \tag{12b}$$

$$\underline{P}_{\mathrm{CHP}} \le P_t^{\mathrm{CHP}} \le \overline{P}_{\mathrm{CHP}} \tag{12c}$$

$$\underline{Q}_{\mathrm{CHP}} \le Q_t^{\mathrm{CHP}} \le \overline{Q}_{\mathrm{CHP}} \tag{12d}$$

$$-\underline{\Delta P}_{\mathrm{CHP}} \le P_t^{\mathrm{CHP}} - P_{t-1}^{\mathrm{CHP}} \le \overline{\Delta P}_{\mathrm{CHP}} \tag{12e}$$

where Eqs. (12a) and (12b) characterize the conversion of natural gas into electricity and heat based on efficiency parameters η and unit conversion factors ρ. Eq. (12c) imposes ramping constraints to limit abrupt changes in CHP output. Eqs. (12d) and (12e) enforce the operational bounds on active and reactive power outputs, while Eq. (12e) defines the maximum allowable reserve capacities that the CHP unit can offer.

Other devices have the same modeling paradigm, and therefore is omitted here for concise.

Device constraints of BESS:

$$E_t^{\mathrm{BE}} = E_{t-1}^{\mathrm{BE}} + \eta_{\mathrm{BE}}^{\mathrm{C}} P_t^{\mathrm{BE,C}} - P_t^{\mathrm{BE,D}} / \eta_{\mathrm{BE}}^{\mathrm{D}} \tag{13a}$$

$$\underline{E}_{\mathrm{BE}} \le E_t^{\mathrm{BE}} \le \overline{E}_{\mathrm{BE}} \tag{13b}$$

$$\underline{P}_{\mathrm{BE,C}} \le P_t^{\mathrm{BE,C}} \le \overline{P}_{\mathrm{BE,C}} \tag{13c}$$

$$\underline{P}_{\mathrm{BE,D}} \le P_t^{\mathrm{BE,D}} \le \overline{P}_{\mathrm{BE,D}} \tag{13d}$$

Eq. (13a) update the energy level of the BESS. Eqs. (13b) and (13c) impose constraints on the maximum and minimum charging and discharging powers. (No additional constraints are required to prevent simultaneous charging and discharging of BESS, since the objective function in (10) is monotonically increasing with respect to related variables).

Power flow constraints:

$$\mathbf{C} \cdot \mathbf{P}_t^F = \left[\boldsymbol{P}_t^B, \boldsymbol{P}_t^{\mathrm{inj}}\right] \tag{14a}$$

$$\mathbf{C} \cdot \boldsymbol{Q}_t^F = \left[0, \boldsymbol{Q}_t^{\mathrm{inj}}\right] \tag{14b}$$

$$\boldsymbol{P}_t^{\mathrm{inj}} = \mathbf{C}_{\mathrm{CHP}} \boldsymbol{P}_t^{\mathrm{CHP}} + \mathbf{C}_{\mathrm{W}} \widetilde{\boldsymbol{P}}_t^{\mathrm{W}} + \mathbf{C}_{\mathrm{FC}} \boldsymbol{P}_t^{\mathrm{FC}} + \mathbf{C}_{\mathrm{BE}} \boldsymbol{P}_t^{\mathrm{BE,D}} - \mathbf{C}_{\mathrm{D}} \boldsymbol{P}_t^{\mathrm{D}} - \mathbf{C}_{\mathrm{EB}} \boldsymbol{P}_t^{\mathrm{EB}} - \mathbf{C}_{\mathrm{BE}} \boldsymbol{P}_t^{\mathrm{BE,C}} - \mathbf{C}_{\mathrm{EL}} \boldsymbol{P}_t^{\mathrm{EL}} \tag{14c}$$

$$\boldsymbol{Q}_t^{\mathrm{inj}} = \mathbf{C}_{\mathrm{CHP}} \boldsymbol{Q}_t^{\mathrm{CHP}} - \mathbf{C}_{\mathrm{D}} \boldsymbol{Q}_t^{\mathrm{D}} \tag{14d}$$

$$\widehat{\mathbf{C}}^{\top}\boldsymbol{v}_t + v_0\mathbf{C}_0 = 2\left(\mathbf{R}\boldsymbol{P}_t^{\mathrm{inj}} + \mathbf{X}\boldsymbol{Q}_t^{\mathrm{inj}}\right) \tag{14e}$$

$$\underline{\boldsymbol{v}} \le \boldsymbol{v} \le \overline{\boldsymbol{v}} \tag{14f}$$

Eqs. (14a) and (14b) establish the relationships between branch-level active and reactive power flows and bus-level power injections, which are defined in (14c) and (14d) via the device-to-bus mapping matrix $\mathbf{C}_*$. Eq. (14e) connects the injected powers to bus voltages by relaxing the phase angle constraints. The squared voltage magnitudes are further bounded within specified limits, as expressed in (14f).

C. Settings for case studies.

The detailed model parameters can be found in the attached Excel file.

In economic dispatch under normal operation, the system diagrams for S1-S3 are illustrated as Fig. S1-S3:

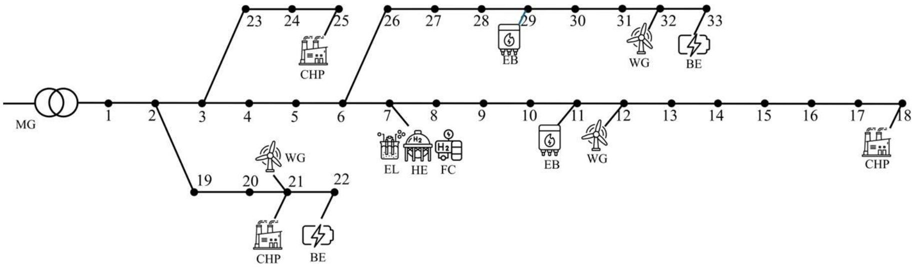


Fig S1. The system diagrams for S1

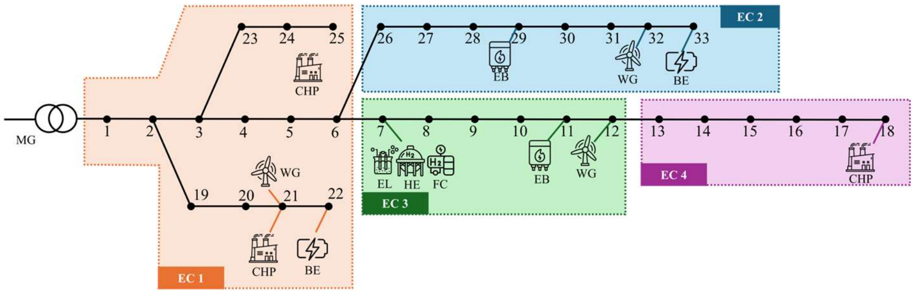


Fig S2. The system diagrams for S2

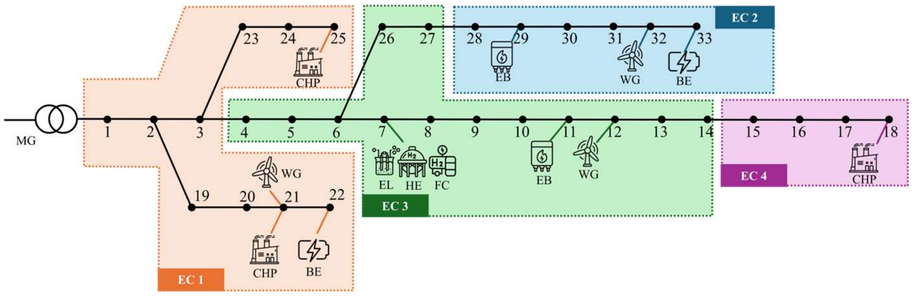


Fig S3. The system diagrams for S3

In resilience enhancement under extreme conditions, the system diagrams without flexible boundaries adjustment are illustrated as Fig. S4:

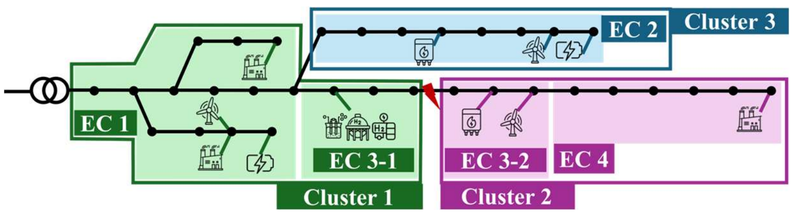
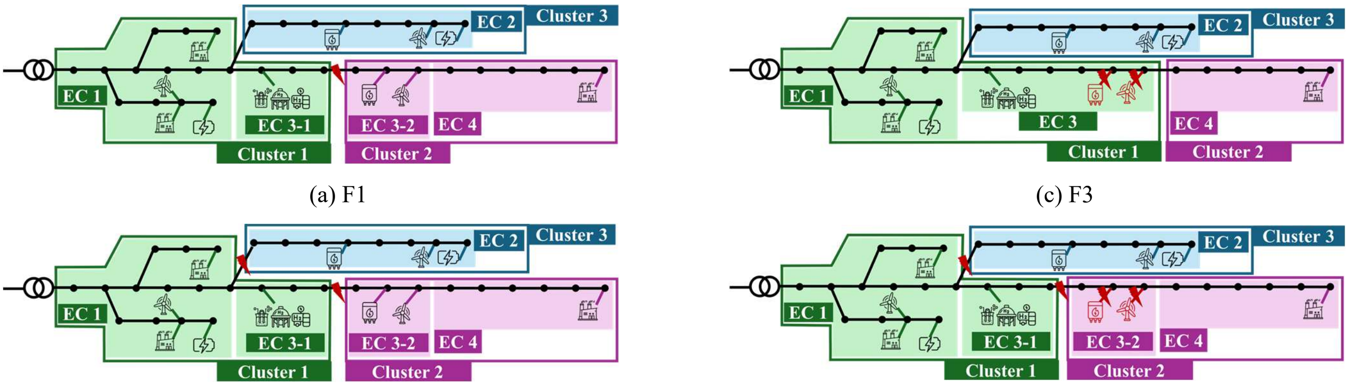


Fig S4. The system diagrams for ECs without boundaries adjustment and clusters

Based on it, we considered four fault conditions with different disturbance types:

- F1: Single line outage. Line 9 in EC3 is disconnected;
- F2: Double line outage. Line 9 in EC3 and line 25 connecting EC1 and EC2 are disconnected;
- F3: Device outage. The wind generator and electric boiler in EC3 are out of service; and
- F4: Combined line & device failures. This scenario combines the above F1-F3 situation.

The cluster formation situations for each fault conditions are illustrated in Fig.S5



(a) F1 (c) F3

(b) F2 (d) F4

Fig S5. The cluster formation situations for each investigated scenario

IV. Additional Figures and Tables

A. The energy flow diagram in EC

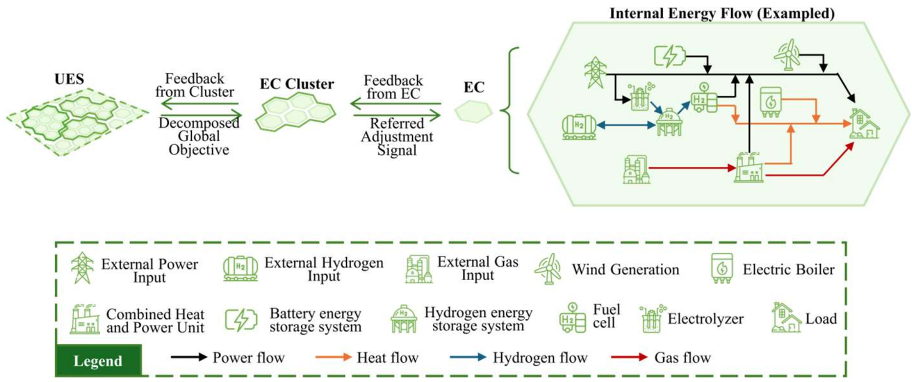


Fig. S6 Energy-flow diagram in an exampled EC and the hierarchy of whole EC framework.

B. The sensitivity analysis in resilience enhancement case

To examine whether the benefits from dynamic boundaries on the resilience conclusion is sensitive to this parameter, we choose the most severe combined fault scenario F4, and compare the load shedding results under different values of λ. The results are summarized in Table S3.

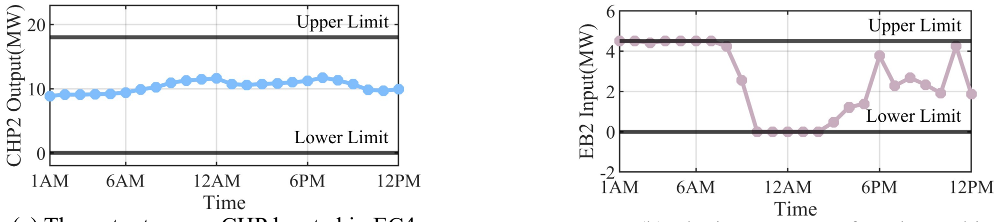
Table S3. Sensitivity analysis of load shedding under different values of λ in F4.

	λ	150	110	100	90	80	70	50	10	1
Shaded Load (MW)	Fixed boundaries	250.99	250.99	250.99	250.99	250.99	202.10	185.52	65.15	0
	Dynamic cluster formation	0	0	0	0	0	0	0	0	0

The results show that, around the baseline value used in the main case study (where $\lambda = 100$), fixed-boundary mode consistently leads to significant load shedding, while the dynamic cluster formation mode does not yield load shedding. For example, when λ varies from 80 to 150, the load shedding under fixed boundaries remains approximately 250.99 MW, and 0 MW under dynamic flexible boundaries. This indicates that the conclusion of the resilience enhancement case is not caused by artificially selected value of λ. It should be noted that, when λ becomes smaller, the load shedding under fixed boundaries gradually decreases. This is reasonable from the engineering meaning of λ. A smaller λ implies a weaker penalty on cross-boundary energy exchange, so the fixed-boundary system is allowed to rely more on external support through the remaining feasible connections. When λ is close to 1, the operation becomes closer to a conventional non-partitioned or weakly partitioned dispatch mode, and the self-balancing requirement of each EC is no longer strongly enforced. Therefore, the 0 MW load shedding under fixed boundaries at a very small λ is achieved by substantially weakening the local self-balancing preference, which deviates from the EC-oriented objective of maintaining localized self-balance. In contrast, the dynamic cluster formation mode eliminates load shedding while preserving the EC-oriented coordination logic, where affected regions are explicitly reorganized into cooperative clusters.

C. Check for violation of devices and system operating constraints

we further illustrate the operating states of representative key devices and branch flows in the F4 combined fault scenario under the dynamic cluster formation mode. Specifically, the outputs of key devices, such as CHP and EB units, as well as the system branch power flows, are checked against their corresponding upper and lower operating limits. (See Fig. S7 and Fig. S8) The results show that these device outputs and branch flows remain within their allowable ranges, indicating that the 0 MW load-shedding result is obtained without violating the modeled device or network constraints.



(a) The output power CHP located in EC4.

(b) The input power of EB located in EC2.

Fig. S7 The energy outputs of key devices.

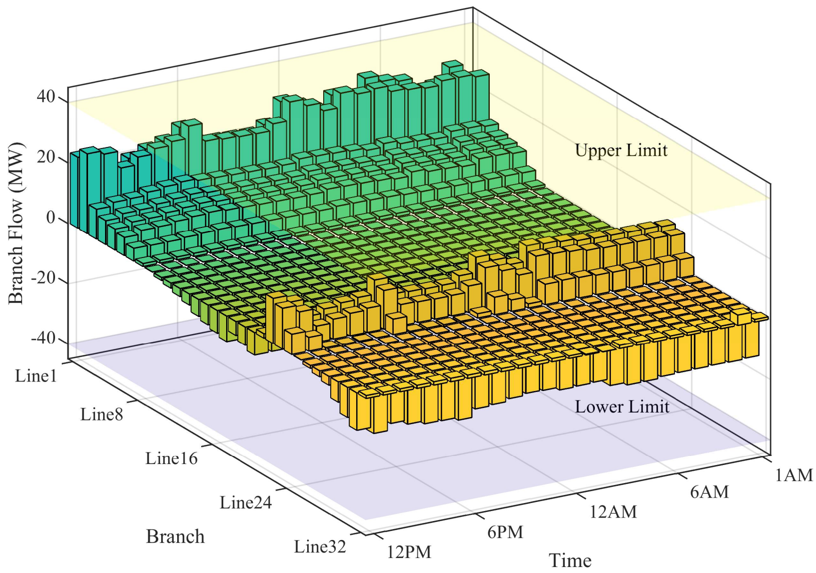


Fig. S8 The power flow of each branch.